

# pid\_datafusion\_exact\_compatibility

*Specialization:* v1

**TL;DR.** The exact observed-law compatibility region for simultaneous relative-NUC and study-exchangeability sensitivity is proved to be the computable zero set  $\mathcal{K}(P) = \{\theta : C_P(\theta) = 0\}$ , equivalently the set of sensitivity pairs admitting one support-respecting common latent causal law. The Lanners-compatible linear endpoints use the factors  $1 - \rho$ ,  $1 + \rho$ ,  $1 - \gamma$ , and  $1 + \gamma$ , with clipping to the known support  $[a, b]$ . This exact criterion differs from the signed-average proxy: in the verified two-stratum law with weights  $4/5$  and  $1/5$ ,  $G_0^{\text{LRVP}}(1, 1) = 0$  but  $C_{P^{\text{can}}}(1, 1) = 9/10$ , so cancellation masks incompatibility. For inference, the proved positive-mass-tie subexperiment has sharp separation rate  $n^{-2\beta/(4\beta+d)}$ , whereas fixed constant paths have the regular  $n^{-1/2}$  Gaussian boundary. Only uniform coverage and uniform level for the proposed cross-fitted numerical directional-delta procedure remain open.

## 1 Project justification

**Gap.** Lanners et al. provide pointwise linear interval restrictions and use a signed expectation as a practical compatibility proxy, with resampling that holds fitted nuisance functions fixed. Those results do not characterize when the support-clipped restrictions are jointly realizable by one latent causal law, and they do not establish tie-inclusive uniform coverage or level after nuisance learning.

**Niche.** The paper isolates the exact no-cancellation compatibility frontier for simultaneous relative-NUC and study-exchangeability sensitivity. It retains the Lanners-compatible linear  $(1 \pm \rho)$  and  $(1 \pm \gamma)$  forms, clips them to  $[a, b]$ , and distinguishes regular pointwise directions from the harder uniform geometry created by positive-mass ties. It does not claim to originate two-parameter sensitivity analysis or randomized-observational data fusion.

**Fill.** The paper proves  $\mathcal{M}(P, \rho, \gamma) \neq \emptyset \iff C_P(\rho, \gamma) = 0$ , verifies the two-stratum cancellation witness with weights  $4/5$  and  $1/5$  and exact violation  $9/10$ , and proves a sharp  $n^{-2\beta/(4\beta+d)}$  minimax separation boundary in a tie subexperiment together with the contrasting root- $n$  Gaussian constant-path power formula. Uniform coverage and uniform level of the specified numerical directional-delta procedure are the only remaining inference questions.

## 2 Related work

Lanners, Rudin, Volfovsky, and Parikh (NeurIPS 2025), including their Project STAR analysis, derive pointwise population interval requirements for simultaneous sensitivity and use  $G_t^{\text{LRVP}}$ , a signed expectation, as a practical proxy while holding fitted nuisances fixed in resampling. The present theorem retains their linear  $(1 \pm \rho)$  and  $(1 \pm \gamma)$  endpoint forms but enforces the known support  $[a, b]$  by clipping and asks the different logical question of existence of one support-respecting common latent causal extension. The nonnegative functional  $C_P$  forbids cross-stratum cancellation: the explicit witness assigns weights  $4/5$  and  $1/5$ , has signed proxy zero for arm zero, and has exact violation  $9/10$ . Yu et al. combine randomized and non-randomized studies with triplet matching and two internal/external-bias sensitivity parameters, but do not give this support-clipped common-extension characterization. Parikh et al. study attribution between external-validity and ignorability failures rather than simultaneous latent-law feasibility. Hong–Li numerical differentiation, Fang–Santos directional inference, Chernozhukov–Lee–Rosen contact selection, and cross-fitting motivate the proposed inference construction; none is claimed here to establish its uniform tie-inclusive coverage or level. The proved minimax subexperiment instead shows why a uniform root- $n$  local-power claim cannot accompany such validity over the frozen primitive class.

## 3 Setup

Target (estimand / structural parameter):  $\mathcal{K}(P)$ . Identifying functional / recovery map:  $\mathcal{K}(P) = \{\theta \in [0, 1]^2 : C_P(\theta) = 0\}$ , where  $C_P(\theta) = \max_{t \in \mathcal{T}} E_{P_X}[(\max\{V_t^-, W_t^-\} - \min\{V_t^+, W_t^+\})_+]$ .

- $n$  — integer;  $\mathbb{N}$ ; index

*Definition.* Sample-size index.

- $d$  — integer;  $\mathbb{N}$ ; dimension

*Definition.* Covariate dimension.

- $\mathcal{X}$  — set;  $\mathbb{R}^d$ ; support

*Definition.* Covariate support.

- $\mathcal{T}$  — set;  $\{0, 1\}$ ; index set

*Definition.* Treatment-state set.

- $t, s, k$  — indices;  $t, k \in \mathcal{T}$ ,  $s \in \{0, 1\}$ ; indices

*Definition.* Treatment and study indices.

- $a, b$  — constants;  $0 < a < b < \infty$ ; support endpoints

*Definition.* Known outcome-support endpoints.

- $\theta = (\rho, \gamma)$  — vector;  $\Theta = [0, 1]^2$ ; index

*Definition.* Sensitivity pair for unmeasured confounding and study exchangeability.

- $\varepsilon, \alpha$  — constants;  $\varepsilon \in (0, 1/2)$ ,  $\alpha \in (0, 1)$ ; constants

*Definition.* Positivity and noncoverage constants.

- $O_i = (X_i, S_i, A_i, Y_i)$  — random vector;  $\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b]$ ; observation

*Definition.* Observed unit.

- $P, P_X, f_X$  — probability laws and density;  $P \in \mathcal{P}(\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b])$ ; observed law

*Definition.* Observed law, its covariate marginal, and its Lebesgue density when it exists.

- $Q$  — probability law;  $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b]^2)$ ; latent law

*Definition.* Latent causal law on  $X, S, A, Y(0), Y(1)$ ; the realized outcome is  $Y(A)$ .

- $\pi, e_t, \mu_{st}$  — functions;  $\pi : \mathcal{X} \rightarrow [0, 1]$ ,  $e_t : \mathcal{X} \times \{0, 1\} \rightarrow [0, 1]$ ,  $\mu_{st} : \mathcal{X} \rightarrow [a, b]$ ; nuisances

*Definition.* Observed nuisances  $\pi(x) = P(S = 1 \mid X = x)$ ,  $e_t(x, s) = P(A = t \mid X = x, S = s)$ , and  $\mu_{st}(x) = E_P[Y \mid X = x, S = s, A = t]$ .

- $c_t, q_{t0}$  — functions;  $\mathcal{X} \rightarrow [a, b]$ ; latent means

*Definition.* Latent means  $c_t(x) = E_Q[Y(t) \mid X = x, S = 0, A = 1 - t]$  and  $q_{t0}(x) = E_Q[Y(t) \mid X = x, S = 0]$ .

- $v_t^\pm, w_t^\pm, H_t^{\text{LRVP}}, G_t^{\text{LRVP}}$  — functions;  $\mathcal{X} \times \Theta \rightarrow \mathbb{R}$ ; comparator functionals

*Definition.* Published un-clipped envelopes, signed overlap, and practical proxy.

- $C_t^\pm, V_t^\pm, W_t^\pm$  — functions;  $\mathcal{X} \times \Theta \rightarrow [a, b]$ ; exact envelopes

*Definition.* Support-clipped missing-arm, trial, and observational-mixture envelopes.

- $C_P, \mathcal{K}(P), \mathcal{M}(P, \rho, \gamma)$  — functional and sets;  $C_P : \Theta \rightarrow [0, \infty)$ ; identified objects

*Definition.* Observed compatibility functional, its zero set, and the compatible latent-law class.

- $m, \nu_m, q_t^*, c_t^*$  — scalar, law, and functions;  $m \in [a, b]$ ,  $\nu_m \in \mathcal{P}([a, b])$ ; extension construction

*Definition.* Endpoint outcome kernel and measurable selector route for a latent extension.

- $P^{\text{can}}$  — probability law;  $\mathcal{P}(\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b])$ ; counterexample

*Definition.* Two-stratum cancellation witness.

- $\beta, B, \mathcal{H}_B^\beta, \underline{f}, \bar{f}, \bar{P}$  — regularity objects;  $\beta > d/2$ ,  $B > 0$ ,  $0 < \underline{f} < \bar{f} < \infty$ ; inference model

*Definition.* Hölder regularity, density bounds, and primitive observed-law class.

- $J, \mathcal{I}_j, p_K, K_n$  — folds and basis;  $J \geq 2$ ,  $K = K_n$ ; learner construction

*Definition.* Cross-fitting partition and normalized tensor-product spline or wavelet basis.

- $\hat{\eta}^{(-j)}, \hat{\pi}^{(-j)}, \hat{e}_t^{(-j)}, \hat{\mu}_{st}^{(-j)}$  — functions;  $\hat{\eta}^{(-j)} = (\hat{\pi}^{(-j)}, \hat{e}^{(-j)}, \hat{\mu}^{(-j)})$ ; learners

*Definition.* Foldwise truncated series-logit and bounded least-squares nuisance fits.

- $h, \hat{\eta}_n, \mathfrak{F}, r_{\pi,n}, r_{e,n}, r_{\mu,n}, F, q, v, A_0, P_n$  — face, fitted vector, map, rates, constants, and law sequence;  $h \in \{V_t^-, V_t^+, W_t^-, W_t^+\}$ ,  $q > 4$ ,  $P_n \in \bar{P}$ ; performance notation

*Definition.* A face functional, its cross-fitted vector, the nested functional map, error-rate labels, entropy-envelope constants, and a law sequence used only by the inference construction or question.

- $p_s, R_{sk}^\mu, R_{sk}^e, \psi_h, \mathcal{S}_n^{\text{act}}$  — scores;  $\mathcal{S}_n^{\text{act}} \subset L_2(P)$ ; orthogonal-score construction

*Definition.* Study probabilities, residual representers, face scores, and tie-retaining active-score collection.

- $\xi_i, \mathbb{G}_n^\xi, \delta_n, \Theta_n, h_n$  — multipliers, process, and sequences;  $E[\xi_i] = 0$ ,  $E[\xi_i^2] = 1$ ,  $\delta_n \downarrow 0$ ,  $h_n \downarrow 0$ ; bootstrap construction

*Definition.* Multiplier process, numerical-delta step, deterministic sensitivity grid, and mesh.

- $\hat{C}_n, \hat{C}_n^\xi, \hat{c}_{1-\alpha}, \hat{\mathcal{K}}_{n,1-\alpha}$  — processes and random set;  $\ell^\infty(\Theta) \times \ell^\infty(\Theta) \times \mathbb{R} \times \mathcal{P}(\Theta)$ ; inference objects

*Definition.* Estimated functional, numerical-delta bootstrap process, critical value, and inverted outer confidence region.

## 4 Assumptions

**Assumption 1** (ass:iid).  $O_1, \dots, O_n \stackrel{\text{iid}}{\sim} P$ .

Standard (*independent sampling*, [8]).

**Assumption 2** (ass:pushforward).  $\text{Law}_Q(X, S, A, Y(A)) = P$ .

Standard (*observed-data pushforward*, [10]).

**Assumption 3** (ass:trial-randomization).  $(Y(0), Y(1)) \perp_Q A \mid (X, S = 1)$ .

Standard (*joint conditional trial randomization*, [2]).

**Assumption 4** (ass:relative-nuc).  $c_t(X) \in [C_t^-(X; \rho), C_t^+(X; \rho)]$   $P_X$ -almost surely for every  $t \in \mathcal{T}$ .

Standard (*linear relative unmeasured-confounding sensitivity model*, [2]).

**Assumption 5** (ass:study-exchangeability).  $q_{t0}(X) \in [V_t^-(X; \gamma), V_t^+(X; \gamma)]$   $P_X$ -almost surely for every  $t \in \mathcal{T}$ .

Standard (*linear study-exchangeability sensitivity model*, [2]).

**Assumption 6** (ass:uniform-study-positivity).  $\varepsilon \leq \pi(X) \leq 1 - \varepsilon$   $P$ -almost surely.

Standard (*uniform study positivity*, [2]).

**Assumption 7** (ass:uniform-treatment-positivity).  $\varepsilon \leq e_t(X, s) \leq 1 - \varepsilon$   $P$ -almost surely for every  $t \in \mathcal{T}$  and  $s \in \{0, 1\}$ .

Standard (*uniform treatment positivity*, [2]).

**Assumption 8** (ass:compact-support).  $\mathcal{X} \subset \mathbb{R}^d$  is compact.

Standard (*compact covariate support*, [8]).

**Assumption 9** (ass:density-bounds).  $\underline{f} \leq f_X(x) \leq \bar{f}$  for Lebesgue-almost every  $x \in \mathcal{X}$ .

Standard (density bounded above and away from zero, [8]).

**Assumption 10** (ass:holder-nuisances).  $\pi \in \mathcal{H}_B^\beta(\mathcal{X})$ ,  $e_t(\cdot, s) \in \mathcal{H}_B^\beta(\mathcal{X})$ , and  $\mu_{st} \in \mathcal{H}_B^\beta(\mathcal{X})$  for every  $t \in \mathcal{T}$  and  $s \in \{0, 1\}$ .

Standard (uniform Hölder nuisance balls, [11]).

## 5 Definitions

**Definition 1** (def:published-endpoints). Defined object.  $v_t^\pm, w_t^\pm, H_t^{\text{LRVP}}, G_t^{\text{LRVP}}$

Construction.  $v_t^-(x; \gamma) = (1 - \gamma)\mu_{1t}(x)$ ,  $v_t^+(x; \gamma) = (1 + \gamma)\mu_{1t}(x)$ ,  $w_t^-(x; \rho) = e_t(x, 0)\mu_{0t}(x) + e_{1-t}(x, 0)(1 - \rho)\mu_{0t}(x)$ ,  $w_t^+(x; \rho) = e_t(x, 0)\mu_{0t}(x) + e_{1-t}(x, 0)(1 + \rho)\mu_{0t}(x)$ ,  $H_t^{\text{LRVP}}(x; \theta) = \min\{v_t^+, w_t^+\} - \max\{v_t^-, w_t^-\}$ , and  $G_t^{\text{LRVP}}(\theta) = E_{P_X}[H_t^{\text{LRVP}}(X; \theta)]$ , for  $x \in \mathcal{X}$  and  $t \in \mathcal{T}$ .

Inputs.  $\pi, e_t, \mu_{st}; P, P_X, f_X; \theta = (\rho, \gamma); t, s, k$ .

**Definition 2** (def:exact-endpoints). Defined object.  $C_t^\pm, V_t^\pm, W_t^\pm$

Construction.  $C_t^-(x; \rho) = \max\{a, (1 - \rho)\mu_{0t}(x)\}$ ,  $C_t^+(x; \rho) = \min\{b, (1 + \rho)\mu_{0t}(x)\}$ ,  $V_t^-(x; \gamma) = \max\{a, (1 - \gamma)\mu_{1t}(x)\}$ ,  $V_t^+(x; \gamma) = \min\{b, (1 + \gamma)\mu_{1t}(x)\}$ , and  $W_t^\pm(x; \rho) = e_t(x, 0)\mu_{0t}(x) + e_{1-t}(x, 0)C_t^\pm(x; \rho)$ , for  $x \in \mathcal{X}$ ,  $t \in \mathcal{T}$ , and  $\theta \in [0, 1]^2$ .

Inputs.  $a, b; \pi, e_t, \mu_{st}; \theta = (\rho, \gamma); t, s, k$ .

**Definition 3** (def:compatibility-functional). Defined object.  $C_P, \mathcal{K}(P)$

Construction.

$C_P(\theta) = \max_{t \in \mathcal{T}} E_{P_X}[(\max\{V_t^-(X; \gamma), W_t^-(X; \rho)\} - \min\{V_t^+(X; \gamma), W_t^+(X; \rho)\})_+]$

and  $\mathcal{K}(P) = \{\theta \in \Theta : C_P(\theta) = 0\}$ .

Inputs.  $P, P_X, f_X; C_t^\pm, V_t^\pm, W_t^\pm; \mathcal{T}; \theta = (\rho, \gamma)$ .

**Definition 4** (def:endpoint-kernel). Defined object.  $\nu_m$

Construction.  $\nu_m = ((b - m)/(b - a))\delta_a + ((m - a)/(b - a))\delta_b$  for every  $m \in [a, b]$ .

Inputs.  $m, \nu_m, q_t^*, c_t^*; a, b$ .

**Definition 5** (def:selector-extension-handle). Defined object.  $q_t^*, c_t^*$

Construction. For each  $t \in \mathcal{T}$ , set  $q_t^*(x; \theta) = [\max\{V_t^-(x; \gamma), W_t^-(x; \rho)\} + \min\{V_t^+(x; \gamma), W_t^+(x; \rho)\}]/2$  and  $c_t^*(x; \theta) = [q_t^*(x; \theta) - e_t(x, 0)\mu_{0t}(x)]/e_{1-t}(x, 0)$ . The construction route copies the observed trial arm- $t$  outcome kernel to  $Y(t)$  under both trial assignments and, in study zero, uses  $\nu_{c_t^*(X)}$  for each missing potential outcome, conditionally independently given  $X, S, A$ .

Inputs.  $C_t^\pm, V_t^\pm, W_t^\pm; \pi, e_t, \mu_{st}; m, \nu_m, q_t^*, c_t^*; \theta = (\rho, \gamma); t, s, k$ .

**Definition 6** (def:latent-class). Defined object.  $\mathcal{M}(P, \rho, \gamma)$

Construction.

$\mathcal{M}(P, \rho, \gamma) = \{Q \in \mathcal{P}(\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b]^2) : \text{ass:pushforward} \wedge \text{ass:trial-randomization} \wedge \text{ass:relative-nuc} \wedge \text{ass:study-exchangeability}\}$

Member properties. **ass:pushforward; ass:trial-randomization; ass:relative-nuc; ass:study-exchangeability.**

**Definition 7** (def:two-stratum-witness). Defined object.  $P^{\text{can}}$

Construction. Let  $\mathcal{X} = \{x_1, x_2\}$ ,  $P^{\text{can}}(X = x_1) = 4/5$ ,  $P^{\text{can}}(X = x_2) = 1/5$ , and  $P^{\text{can}}(S = s, A = t \mid X = x_j) = 1/4$ . Set  $a = 1/4$ ,  $b = 8$ ,  $\theta = (1, 1)$ ,  $\mu_{00}(x_1) = \mu_{00}(x_2) = 4$ ,  $\mu_{10}(x_1) = 1/2$ ,  $\mu_{10}(x_2) = 4$ , and  $\mu_{01}(x_j) = \mu_{11}(x_j) = 1$ . For every  $j \in \{1, 2\}$ ,  $s \in \{0, 1\}$ , and  $t \in \mathcal{T}$ , let  $Y \mid X = x_j, S = s, A = t \sim \nu_{\mu_{st}(x_j)}$ .

Inputs.  $P^{\text{can}}; a, b; \pi, e_t, \mu_{st}; m, \nu_m, q_t^*, c_t^*; \theta = (\rho, \gamma)$ .

**Definition 8** (def:holder-ball). Defined object.  $\mathcal{H}_B^\beta(\mathcal{X})$

Construction.

$$\mathcal{H}_B^\beta(\mathcal{X}) = \{g : \mathcal{X} \rightarrow \mathbb{R} : \|g\|_{C^\beta(\mathcal{X})} \leq B\}$$

Inputs.  $\mathcal{X}; \beta, B, \mathcal{H}_B^\beta, \underline{f}, \bar{f}, \bar{\mathcal{P}}$ .

**Definition 9** (def:primitive-class). Defined object.  $\bar{\mathcal{P}}$

Construction.

$$\bar{\mathcal{P}} = \{P \in \mathcal{P}(\mathcal{X} \times \{0, 1\} \times \mathcal{T} \times [a, b]) : \text{ass:compact-support} \wedge \text{ass:density-bounds} \wedge \text{ass:uniform-study-positivity} \wedge \text{ass:uniform-treatment-positivity} \wedge \text{ass:holder-nuisances}\}$$

Member properties. *ass:compact-support; ass:density-bounds; ass:uniform-study-positivity; ass:uniform-treatment-positivity; ass:holder-nuisances.*

**Definition 10** (def:inference-handle). Defined object.  $\hat{C}_n, \hat{C}_n^\xi, \hat{c}_{1-\alpha}, \hat{\mathcal{K}}_{n,1-\alpha}$

Construction. Use a normalized tensor-product spline or compactly supported wavelet basis  $p_K(x)$  of order at least  $\lceil \beta \rceil$ , with  $K = K_n$ , and fixed  $J$ -folds. On training complements fit truncated series logits  $\hat{\pi}^{(-j)}, \hat{e}_k^{(-j)}$  and bounded series least squares  $\hat{\mu}_{sk}^{(-j)}$ . Require for this construction  $K_n \log n/n = o(1)$ ,  $K_n^{-\beta/d} = o(n^{-1/4})$ , foldwise uniform errors  $r_{\pi,n}, r_{e,n}, r_{\mu,n}$ , and  $r_{\mu,n}(r_{e,n} + r_{\pi,n}) = o(n^{-1/2})$ . Put  $p_s(x) = \pi(x)^s [1 - \pi(x)]^{1-s}$ ,  $R_{sk}^\mu = 1\{S = s, A = k\}[Y - \mu_{sk}(X)]/[p_s(X)e_k(X, s)]$ , and  $R_{sk}^e = 1\{S = s\}[1\{A = k\} - e_k(X, s)]/p_s(X)$ . For any face  $h \in \{V_t^-, V_t^+, W_t^-, W_t^+\}$ , define

$$\psi_h(O; \theta) = h(X; \theta) - E_P h(X; \theta) + \sum_{s,k} \partial_{\mu_{sk}} h(X; \theta) R_{sk}^\mu + \sum_{s,k} \partial_{e_k(\cdot, s)} h(X; \theta) R_{sk}^e$$

, with branchwise directional derivatives of the displayed clipped formula; the  $\pi$ -derivative is zero. Let  $\mathcal{S}_n^{\text{act}}$  collect centered face scores attaining a lower maximum, upper minimum, positive-part contact, or outer arm maximum at every  $\theta \in \Theta_n$ , retaining all ties. Its sufficient performance conditions are an envelope  $F$  with  $\sup_{P \in \bar{\mathcal{P}}} E_P F^q < \infty$  for some  $q > 4$ ,  $\log N(\eta, \mathcal{S}_n^{\text{act}}, L_2(P)) \leq v \log(A_0/\eta)$  uniformly in  $P$  and  $0 < \eta \leq 1$ , and a fixed studentization floor for nonzero-variance scores; zero-variance scores are deterministically zero. Let  $\hat{C}_n(\theta)$  be the cross-fitted empirical face vector passed through the lower-max, upper-min, positive-part, expectation, and outer-max map  $\mathfrak{F}$ . With independent mean-zero unit-variance multipliers, put  $\mathbb{G}_n^\xi = n^{-1/2} \sum_i \xi_i \hat{\psi}_i$ , choose  $\delta_n \downarrow 0$  with  $n^{-1/2}/\delta_n \rightarrow 0$  and  $(r_{\pi,n} + r_{e,n} + r_{\mu,n})/\delta_n \rightarrow 0$ , and set

$$\hat{C}_n^\xi = [\mathfrak{F}(\hat{\eta}_n + \delta_n \mathbb{G}_n^\xi) - \mathfrak{F}(\hat{\eta}_n)]/\delta_n$$

. Take deterministic  $\Theta_n \subset [0, 1]^2$  with mesh  $h_n \rightarrow 0$  and interpolation error  $o(n^{-1/2})$ . Let  $\hat{c}_{1-\alpha}$  be the conditional  $(1 - \alpha)$ -quantile of  $\sup_{\theta \in \Theta_n} \hat{C}_n^\xi(\theta)$ , and define

$$\hat{\mathcal{K}}_{n,1-\alpha} = \{\theta \in \Theta_n : \sqrt{n} \hat{C}_n(\theta) \leq \hat{c}_{1-\alpha}\}$$

, enlarged by measurable closed grid cells in  $\Theta$ . Hong-Li supplies the finite-difference derivative, Fang-Santos the directional map, Chernozhukov-Lee-Rosen contact selection, and cross-fitting nuisance linearization.

Inputs.  $O_i = (X_i, S_i, A_i, Y_i); J, \mathcal{I}_j, p_K, K_n; \hat{\eta}^{(-j)}, \hat{\pi}^{(-j)}, \hat{e}_t^{(-j)}, \hat{\mu}_{st}^{(-j)}; p_s, R_{sk}^\mu, R_{sk}^e, \psi_h, \mathcal{S}_n^{\text{act}}; \xi_i, \mathbb{G}_n^\xi, \delta_n, \Theta_n, h_n; \varepsilon, \alpha$ .

## 6 Main results

**Lemma 1** (lem:observational-mixture-identity). Every  $Q$  satisfying *ass:pushforward* obeys the weighted observational-mixture identity

$$(1 - \pi(X)) [q_{t0}(X) - e_t(X, 0) \mu_{0t}(X) - e_{1-t}(X, 0) c_t(X)] = 0$$

$P_X$ -almost surely for each  $t \in \mathcal{T}$ . Consequently, the unweighted identity holds  $P_X$ -almost surely on  $\{x : \pi(x) < 1\}$ .

*Proof.* Fix  $t \in \mathcal{T}$ , and define the conditional discrepancy

$$D_t(x) = q_{t0}(x) - e_t(x, 0)\mu_{0t}(x) - e_{1-t}(x, 0)c_t(x).$$

All conditional expectations may be represented by arbitrary measurable versions. Multiplication by the observational-study probability makes the asserted equality invariant to the values of those versions outside the support of  $X \mid S = 0$ . By conditional iterated expectation under  $Q$ , and because  $A \in \{t, 1-t\}$ ,

$$\begin{aligned} q_{t0}(X) &= E_Q[Y(t) \mid X, S = 0] \\ &= Q(A = t \mid X, S = 0)E_Q[Y(t) \mid X, S = 0, A = t] \\ &\quad + Q(A = 1 - t \mid X, S = 0)E_Q[Y(t) \mid X, S = 0, A = 1 - t] \end{aligned}$$

almost surely on  $\{S = 0\}$ . Assumption `ass:pushforward` identifies the two conditional assignment probabilities with  $e_t(X, 0)$  and  $e_{1-t}(X, 0)$ , respectively. On  $\{S = 0, A = t\}$ , consistency of the realized outcome gives  $Y(t) = Y(A)$ , and the same pushforward assumption identifies the corresponding conditional mean with  $\mu_{0t}(X)$ . The other conditional mean is  $c_t(X)$  by definition. Hence  $D_t(X) = 0$   $Q$ -almost surely on  $\{S = 0\}$ . The pushforward also gives  $Q(S = 0 \mid X) = P(S = 0 \mid X) = 1 - \pi(X)$ . Therefore

$$E_{P_X}[(1 - \pi(X))|D_t(X)|] = E_Q[1\{S = 0\}|D_t(X)|] = 0.$$

The integrand is nonnegative, so  $(1 - \pi(X))D_t(X) = 0$   $P_X$ -almost surely. On  $\{x : \pi(x) < 1\}$ , the factor  $1 - \pi(x)$  is strictly positive, and division yields  $D_t(x) = 0$   $P_X$ -almost surely there. Since  $t$  was arbitrary and  $\mathcal{T}$  is finite, the asserted conclusion holds for every  $t \in \mathcal{T}$ .  $\square$

*Justification.* The mixture equality follows from the observed-data pushforward and conditional iterated expectation; it is not an additional causal restriction. *Gap.* A law-level bridge is needed to translate the missing-arm mean restriction into an interval for the observational-study potential-outcome mean. *Consumer.* The positive-violation proposition uses the bridge to map  $c_t \in [C_t^-, C_t^+]$  into  $q_{t0} \in [W_t^-, W_t^+]$ .

**Proposition 1** (`prop:positive-violation-infeasibility`). *Under `ass:uniform-study-positivity` and `ass:uniform-treatment-positivity`,  $C_P(\theta) > 0$  implies  $\mathcal{M}(P, \rho, \gamma) = \emptyset$ . Indeed, for any  $Q \in \mathcal{M}(P, \rho, \gamma)$ , the weighted observational-mixture identity may be divided by  $1 - \pi(X) \geq \varepsilon$ . Since  $c_t \in [C_t^-, C_t^+]$ , it follows that  $q_{t0} \in [W_t^-, W_t^+]$ , while study exchangeability gives  $q_{t0} \in [V_t^-, V_t^+]$ . Thus every nonnegative conditional violation is zero almost surely, contradicting  $C_P(\theta) > 0$ .*

*Proof.* Suppose, toward a contradiction, that  $C_P(\theta) > 0$  and that some  $Q \in \mathcal{M}(P, \rho, \gamma)$  exists. By `def:latent-class`, this law satisfies `ass:pushforward`, `ass:relative-nuc`, and `ass:study-exchangeability`. By `lem:observational-mixture-identity`, for each  $t \in \mathcal{T}$ ,

$$(1 - \pi(X))\{q_{t0}(X) - e_t(X, 0)\mu_{0t}(X) - e_{1-t}(X, 0)c_t(X)\} = 0$$

almost surely. Assumption `ass:uniform-study-positivity` gives  $1 - \pi(X) \geq \varepsilon > 0$ , so division yields

$$q_{t0}(X) = e_t(X, 0)\mu_{0t}(X) + e_{1-t}(X, 0)c_t(X) \quad P_X\text{-almost surely.}$$

Moreover, `ass:uniform-study-positivity` and `ass:uniform-treatment-positivity` imply

$$P(S = 0, A = k \mid X) = (1 - \pi(X))e_k(X, 0) \geq \varepsilon^2$$

for every  $k \in \mathcal{T}$ , so the arm-specific conditional means are defined on the relevant  $P_X$ -support. The binary-treatment probabilities are nonnegative and sum to one. Assumption `ass:relative-nuc`, together with `def:exact-endpoints`, therefore implies endpoint by endpoint that

$$W_t^-(X; \rho) \leq q_{t0}(X) \leq W_t^+(X; \rho) \quad P_X\text{-almost surely.}$$

Assumption `ass:study-exchangeability` simultaneously gives

$$V_t^-(X; \gamma) \leq q_{t0}(X) \leq V_t^+(X; \gamma) \quad P_X\text{-almost surely.}$$

Define the proof intermediates

$$L_t(X; \theta) = \max\{V_t^-(X; \gamma), W_t^-(X; \rho)\}, \quad U_t(X; \theta) = \min\{V_t^+(X; \gamma), W_t^+(X; \rho)\}.$$

The two interval inclusions give  $L_t(X; \theta) \leq q_{t0}(X) \leq U_t(X; \theta)$  almost surely. Consequently,  $(L_t - U_t)_+ = 0$  almost surely and  $E_{P_X}[(L_t - U_t)_+] = 0$  for both treatment arms. Definition def:compatibility-functional now gives  $C_P(\theta) = 0$ , contradicting  $C_P(\theta) > 0$ . Hence  $\mathcal{M}(P, \rho, \gamma) = \emptyset$ .  $\square$

*Justification.* This necessity result turns any positive no-cancellation gap into an observed-law certificate that no common latent extension exists. *Gap.* It establishes infeasibility without appealing to a signed average and isolates the converse construction as a separate theorem-level task. *Consumer.* It certifies the emptiness of the latent class in the two-stratum witness and supports practical rejection of incompatible sensitivity pairs.

**Proposition 2** (prop:two-stratum-cancellation). *For  $P^{\text{can}}$  at  $\theta = (1, 1)$ , all means lie in  $[1/4, 8]$ ,  $\pi(x_j) = e_t(x_j, s) = 1/2$ , and all observed kernels are valid. For arm 0, the published un-clipped intervals are  $[0, 1]$  and  $[2, 6]$  at  $x_1$ , giving  $H_0^{\text{LRVP}}(x_1) = -1$ , and  $[0, 8]$  and  $[2, 6]$  at  $x_2$ , giving  $H_0^{\text{LRVP}}(x_2) = 4$ . Hence  $G_0^{\text{LRVP}} = (4/5)(-1) + (1/5)4 = 0$ . For arm 1, the published intervals are  $[0, 2]$  and  $[1/2, 3/2]$  at both strata, so  $H_1^{\text{LRVP}} = G_1^{\text{LRVP}} = 1$ . The support-clipped arm-zero intervals are  $V_0 = [1/4, 1]$ ,  $C_0 = [1/4, 8]$ , and  $W_0 = [17/8, 6]$  at  $x_1$ , and  $V_0 = [1/4, 8]$ ,  $C_0 = [1/4, 8]$ , and  $W_0 = [17/8, 6]$  at  $x_2$ . For arm 1, they are  $V_1 = [1/4, 2]$ ,  $C_1 = [1/4, 2]$ , and  $W_1 = [5/8, 3/2]$  at both strata. Thus the only conditional violation is  $9/8$  at  $x_1$  for arm 0, so  $C_{P^{\text{can}}}(1, 1) = 9/10$  and  $\mathcal{M}(P^{\text{can}}, 1, 1) = \emptyset$  by Proposition 1.*

*Proof.* Under  $P^{\text{can}}$ , every listed mean belongs to  $[1/4, 8]$ . For  $m \in [1/4, 8]$ , def:endpoint-kernel gives  $\nu_m$  nonnegative weights summing to one, support contained in  $\{1/4, 8\}$ , and mean  $m$ ; hence every displayed observed outcome kernel is valid. Moreover,

$$\pi(x_j) = \sum_{t \in \mathcal{T}} P^{\text{can}}(S = 1, A = t \mid X = x_j) = \frac{1}{2},$$

and, for every  $s, t, j$ ,

$$e_t(x_j, s) = \frac{P^{\text{can}}(S = s, A = t \mid X = x_j)}{P^{\text{can}}(S = s \mid X = x_j)} = \frac{1/4}{1/2} = \frac{1}{2}.$$

At  $\rho = \gamma = 1$ , the published endpoints in def:published-endpoints reduce to

$$v_t^- = 0, \quad v_t^+ = 2\mu_{1t}, \quad w_t^- = \frac{1}{2}\mu_{0t}, \quad w_t^+ = \frac{3}{2}\mu_{0t}.$$

For arm 0, the trial and observational intervals at  $x_1$  are therefore  $[0, 1]$  and  $[2, 6]$ , so

$$H_0^{\text{LRVP}}(x_1; (1, 1)) = \min\{1, 6\} - \max\{0, 2\} = -1.$$

At  $x_2$  they are  $[0, 8]$  and  $[2, 6]$ , so  $H_0^{\text{LRVP}}(x_2; (1, 1)) = 6 - 2 = 4$ . It follows that

$$G_0^{\text{LRVP}}(1, 1) = \frac{4}{5}(-1) + \frac{1}{5}(4) = 0.$$

For arm 1, both strata have published intervals  $[0, 2]$  and  $[1/2, 3/2]$ , whence  $H_1^{\text{LRVP}} = 1$  pointwise and  $G_1^{\text{LRVP}}(1, 1) = 1$ . For the support-clipped endpoints in def:exact-endpoints, the equalities  $1 - \rho = 1 - \gamma = 0$  give  $C_t^- = V_t^- = 1/4$ , whereas  $C_t^+ = \min\{8, 2\mu_{0t}\}$  and  $V_t^+ = \min\{8, 2\mu_{1t}\}$ . Thus, for arm 0,

$$V_0(x_1) = [1/4, 1], \quad V_0(x_2) = [1/4, 8], \quad C_0(x_j) = [1/4, 8].$$

Since  $e_0 = e_1 = 1/2$  and  $\mu_{00} = 4$ , both strata have

$$W_0(x_j) = \left[ \frac{1}{2}(4) + \frac{1}{2} \left( \frac{1}{4} \right), \frac{1}{2}(4) + \frac{1}{2}(8) \right] = [17/8, 6].$$

For arm 1, both  $\mu_{01}$  and  $\mu_{11}$  equal one, and hence

$$V_1(x_j) = C_1(x_j) = [1/4, 2], \quad W_1(x_j) = \left[ \frac{1}{2}(1) + \frac{1}{2} \left( \frac{1}{4} \right), \frac{1}{2}(1) + \frac{1}{2}(2) \right] = [5/8, 3/2].$$

The positive gap between the maximum lower endpoint and the minimum upper endpoint is therefore  $17/8 - 1 = 9/8$  for arm 0 at  $x_1$ , zero for arm 0 at  $x_2$ , and zero for arm 1 at either stratum. Definition def:compatibility-functional yields

$$C_{P^{\text{can}}}(1, 1) = \max \left\{ \frac{4}{5} \frac{9}{8} + \frac{1}{5}(0), 0 \right\} = \frac{9}{10}.$$

Finally, since  $\varepsilon < 1/2$ , the equalities  $\pi = e_t = 1/2$  satisfy both uniform positivity assumptions. Proposition prop:positive-violation-infeasibility therefore applies and gives  $\mathcal{M}(P^{\text{can}}, 1, 1) = \emptyset$ , even though the published arm-zero signed average equals zero.  $\square$

*Justification.* The completely specified observed law verifies numerically that signed covariate-stratum cancellation can mask exact incompatibility. *Gap.* A concrete witness is needed to show that the distinction between the published proxy and the exact functional is substantive rather than terminological. *Consumer.* Applied readers can use the witness as a diagnostic warning:  $G_0^{\text{LRVP}} = 0$  can coexist with  $C_P = 9/10$  and an empty compatible class.

**Open-ended Question 1** (oeq:uniform-contact-inference). *Under  $P \in \overline{\mathcal{P}}$ , ass:iid, and the construction-performance conditions in def:inference-handle, can the numerical directional-delta contact-set multiplier procedure attain*

$$\liminf_{n \rightarrow \infty} \inf_{P \in \overline{\mathcal{P}}} P(\mathcal{K}(P) \subseteq \widehat{\mathcal{K}}_{n, 1-\alpha}) \geq 1 - \alpha$$

and yield uniformly level- $\alpha$  tests of  $C_P(\theta) = 0$ , including positive- $P_X$ -mass endpoint and arm ties?

The proved thm:tie-local-separation-obstruction shows that, in the canonical positive-mass endpoint-and-arm-tie subexperiment, the sharp uniform separation order is  $n^{-2\beta/(4\beta+d)}$ , while lem:regular-constant-path-power gives the pointwise  $n^{-1/2}$  Gaussian boundary along fixed constant directions. Thus no uniformly level procedure can have a nontrivial worst-case power curve indexed only by  $\sqrt{n}C_P(\theta)$  over the full tie-inclusive class. The additional lem:scalar-face-tangent-nonfactorization proves that the displayed scalar multiplier input is insufficient to determine the tie derivative: two paths with identical integrated face derivatives have compatibility derivatives 0 and  $1/4$ . Uniform coverage and uniform null level remain logically possible only after a selector-complete function-valued procedure and its missing uniform approximation and inversion arguments are supplied.

*Justification.* After the negative local-separation resolution, the honest residual question is uniform coverage and test level for the specific tie-retaining numerical directional-delta multiplier procedure. *Gap.* No current theorem in the core supplies the tie-uniform bootstrap approximation and inversion argument required for those guarantees. *Consumer.* Resolving this question would determine whether the computable population frontier is accompanied by a uniformly valid confidence region and null test.

**Lemma 2** (lem:standard-borel-conditional-kernel). *Let  $Z$  and  $W$  be random elements, with  $W$  taking values in a standard-Borel space. There exists a probability kernel  $z \mapsto \kappa(z, \cdot)$  that is a regular conditional law of  $W$  given  $Z = z$ , and it is unique for  $\text{Law}(Z)$ -almost every  $z$ .*

**Lemma 3** (lem:tie-subexperiment-minimax-boundary). *The primitive conditions admit the following specialization for every  $d \geq 1$  and  $\beta > d/2$ : take  $\mathcal{X} = [0, 1]^d$ ,  $X$  uniform,  $S$  and  $A$  independent fair Bernoulli variables, and an interior baseline mean  $m_0 = (a + b)/2$ , with  $B$  and the density bounds chosen to contain this submodel. Put  $Z = (Y - a)/(b - a)$ . For a function  $u : [0, 1]^d \rightarrow [-1/4, 1/4]$ , let  $P_u$  have  $Z$  Bernoulli with conditional success probability  $1/2 + u(X)$  when  $S = 1, A = 0$ , and success probability  $1/2$  in every other study-arm cell. Restrict  $u$  to a fixed bounded  $C^\beta$ -ball small enough that the resulting outcome regressions lie in  $\mathcal{H}_B^\beta(\mathcal{X})$ . Then*

$$C_{P_u}(0, 0) = (b - a)\|u\|_{L_1([0, 1]^d)}.$$

Let  $r_n = n^{-2\beta/(4\beta+d)}$ . For every  $\eta > 0$ , there is  $c_\eta > 0$  such that every test  $\varphi_n$  with  $E_{P_0}\varphi_n \leq \alpha + o(1)$  has, for some  $P_{u_n}$ ,

$$C_{P_{u_n}}(0, 0) \asymp c_\eta r_n, \quad E_{P_{u_n}}\varphi_n \leq \alpha + \eta + o(1).$$

There likewise exist alternatives with  $C_{P_{u_n}}(0, 0) \asymp n^{-1/2}$  and power at most  $\alpha + o(1)$ . Conversely, let  $p_1, \dots, p_K$  be an orthonormal piecewise-polynomial basis of order at least  $\lceil \beta \rceil$ , put  $K \asymp n^{2d/(4\beta+d)}$ , define

$$R_i = 4\{1\{S_i = 1, A_i = 0\} - 1\{S_i = 0, A_i = 0\}\}Z_i, \quad \widehat{Q}_K = \frac{1}{n(n-1)} \sum_{i \neq j} R_i R_j \sum_{k=1}^K p_k(X_i) p_k(X_j),$$



and reject for  $\widehat{Q}_K > L_\alpha \sqrt{K}/n$ , where  $L_\alpha$  is a sufficiently large constant. This test has level at most  $\alpha$  under  $P_0$  and power tending to one uniformly over the displayed Hölder submodel whenever  $C_{P_u}(0,0)/r_n \rightarrow \infty$ . Thus  $r_n$  is the sharp minimax separation order in this positive-mass arm-and-endpoint-tie subexperiment.

*Proof.* We first verify membership and compute the target. The chosen study and treatment probabilities equal  $1/2$ , so both positivity assumptions hold for every  $\varepsilon < 1/2$ . The covariate density is one. The endpoint law of  $Y$  is  $\nu_m$  with  $m = m_0 + (b-a)u(x)$  in the cell  $S = 1, A = 0$  and  $m = m_0$  elsewhere. Choosing the radius of the  $C^\beta$ -ball for  $u$  sufficiently small keeps every mean in  $[a, b]$  and every nuisance in the prescribed Hölder ball. Thus these laws form an admissible specialization of  $\overline{\mathcal{P}}$ .

At  $\rho = \gamma = 0$ , support clipping is inactive because every conditional mean belongs to  $[a, b]$ , and the binary-treatment identity  $e_0(x, s) + e_1(x, s) = 1$  gives

$$C_t^-(x; 0) = C_t^+(x; 0) = \mu_{0t}(x), \quad V_t^-(x; 0) = V_t^+(x; 0) = \mu_{1t}(x), \quad W_t^-(x; 0) = W_t^+(x; 0) = \mu_{0t}(x).$$

Consequently, the conditional violation for arm  $t$  is  $|\mu_{1t}(x) - \mu_{0t}(x)|$ . Only arm zero differs in the submodel, whence

$$C_{P_u}(0, 0) = (b-a) \int_{[0,1]^d} |u(x)| dx.$$

All four faces and the positive part tie on the whole covariate support at  $u = 0$ , so the tie has positive, indeed unit,  $P_X$ -mass.

We next prove the lower bound by an explicit mixture. Choose a nonzero nonnegative  $C^\infty$  function  $\psi$ , supported in the interior of  $[0, 1]^d$ , with  $\|\psi\|_\infty \leq 1$ . For a small bandwidth  $h$ , pack  $M \asymp h^{-d}$  disjoint cubes of side  $h$  into  $[0, 1]^d$ , with lower corners  $x_1, \dots, x_M$ . For  $\omega = (\omega_1, \dots, \omega_M) \in \{-1, 1\}^M$ , set

$$u_\omega(x) = ch^\beta \sum_{j=1}^M \omega_j \psi\left(\frac{x - x_j}{h}\right).$$

The supports are disjoint. The scaling identity for derivatives and the Hölder remainder show directly that  $\|u_\omega\|_{C^\beta} \leq C_\psi c$ , uniformly in  $h$  and  $\omega$ . Taking  $c$  small therefore places every alternative in the submodel and in  $[-1/4, 1/4]$ . Moreover,

$$\|u_\omega\|_1 = ch^\beta M h^d \int |\psi(z)| dz \asymp ch^\beta.$$

Let  $L_\omega = dP_{u_\omega}/dP_0$  for one observation. Conditional on  $G = 1\{S = 1, A = 0\}$ , the Bernoulli likelihood ratio is  $1 + 2(2Z - 1)u_\omega(X)$ , and it is one when  $G = 0$ . Since  $P_0(G = 1) = 1/4$ ,  $E_{P_0}(2Z - 1) = 0$ , and  $(2Z - 1)^2 = 1$ , direct multiplication gives

$$E_{P_0}[L_\omega L_{\omega'}] = 1 + \langle u_\omega, u_{\omega'} \rangle_2.$$

Let  $\overline{P}_n = 2^{-M} \sum_\omega P_{u_\omega}^{\otimes n}$ . Orthogonality of the disjoint bumps gives, for a constant  $\lambda = c^2 \int \psi^2$ ,

$$\langle u_\omega, u_{\omega'} \rangle_2 = \lambda h^{2\beta+d} \sum_{j=1}^M \omega_j \omega'_j.$$

Using  $(1+x)^n \leq e^{nx}$ , averaging independently over  $\omega$  and  $\omega'$ , and using  $\log \cosh z \leq z^2/2$ , we obtain

$$1 + \chi^2(\overline{P}_n, P_0^{\otimes n}) \leq \{\cosh(n\lambda h^{2\beta+d})\}^M \leq \exp\left(\frac{Mn^2\lambda^2 h^{4\beta+2d}}{2}\right) \leq \exp(Cc^4 n^2 h^{4\beta+d}).$$

Take  $h \asymp n^{-2/(4\beta+d)}$ . Then the last exponent is bounded by a constant multiple of  $c^4$ , while every alternative has  $C_{P_{u_\omega}}(0, 0) \asymp c(b-a)r_n$ . By choosing  $c = c_\eta$  sufficiently small, the total-variation inequality

$$\|\overline{P}_n - P_0^{\otimes n}\|_{\text{TV}} \leq \sqrt{\chi^2(\overline{P}_n, P_0^{\otimes n})/2}$$

makes this distance at most  $\eta + o(1)$ . Therefore

$$2^{-M} \sum_\omega E_{P_{u_\omega}} \varphi_n \leq E_{P_0} \varphi_n + \eta + o(1) \leq \alpha + \eta + o(1),$$

so at least one mixture component has the asserted low power. If instead  $h \asymp n^{-1/(2\beta)}$ , then the signal is of order  $n^{-1/2}$  and  $n^2 h^{4\beta+d} = n^{-d/(2\beta)} \rightarrow 0$ . The same calculation gives total variation tending to zero and hence power at most  $\alpha + o(1)$  for at least one root- $n$  alternative.

It remains to prove attainability of the rate. Define the projection kernel  $K_K(x, x') = \sum_{k=1}^K p_k(x)p_k(x')$ . The inverse-cell weight in  $R_i$  gives

$$E_{P_u}[R_i | X_i = x] = 4 \left\{ \frac{1}{4} \left( \frac{1}{2} + u(x) \right) - \frac{1}{4} \left( \frac{1}{2} \right) \right\} = u(x).$$

Independence of two distinct observations therefore yields

$$E_{P_u}[\widehat{Q}_K] = \sum_{k=1}^K \langle u, p_k \rangle_2^2 = \|\Pi_K u\|_2^2.$$

A piecewise Taylor polynomial of degree  $\lfloor \beta \rfloor$  on cubes of side proportional to  $K^{-1/d}$  has pointwise remainder at most  $CK^{-\beta/d}$  by the defining Hölder inequality. Since the orthogonal projection is the best  $L_2$  approximation,

$$\|u - \Pi_K u\|_2 \leq CK^{-\beta/d}.$$

Also  $|R_i| \leq 4$ . Writing the order-two statistic as its first projection plus a degenerate remainder and expanding the variance gives

$$\text{Var}_{P_u}(\widehat{Q}_K) \leq C \left\{ \frac{\|\Pi_K u\|_2^2}{n} + \frac{E[K_K(X_1, X_2)^2]}{n^2} \right\} = C \left\{ \frac{\|\Pi_K u\|_2^2}{n} + \frac{K}{n^2} \right\}.$$

The equality uses orthonormality twice:  $E[K_K(X_1, X_2)^2] = K$ . Under  $u = 0$ , Chebyshev's inequality thus makes the rejection probability at most  $C/L_\alpha^2$ , so a fixed sufficiently large  $L_\alpha$  gives level at most  $\alpha$ . On a unit-volume cube,  $\|u\|_2 \geq \|u\|_1$ . With  $K \asymp n^{2d/(4\beta+d)}$ ,

$$K^{-\beta/d} \asymp r_n, \quad \frac{\sqrt{K}}{n} \asymp r_n^2.$$

Hence  $\|u\|_1/r_n \rightarrow \infty$  implies  $E\widehat{Q}_K/(\sqrt{K}/n) \rightarrow \infty$ , and the displayed variance bound plus Chebyshev's inequality gives rejection probability tending to one uniformly. This proves the matching upper bound and the lemma.  $\square$

**Lemma 4** (lem:regular-constant-path-power). *In the canonical subexperiment of lem:tie-subexperiment-minimax-boundary, fix  $\kappa \in \mathbb{R}$  and take  $u_n(x) = \kappa/[(b-a)\sqrt{n}]$ . Then  $C_{P_{u_n}}(0, 0) = |\kappa|/\sqrt{n}$ . Define*

$$D_i = 4(b-a)\{1\{S_i = 1, A_i = 0\} - 1\{S_i = 0, A_i = 0\}\} \left( Z_i - \frac{1}{2} \right), \quad \sigma^2 = 2(b-a)^2.$$

*Under  $P_{u_n}$ ,  $\sqrt{n}n^{-1}\sum_i D_i/\sigma$  converges to  $N(\kappa/\sigma, 1)$ . If  $\Phi$  is the standard-normal distribution function and  $z_p = \Phi^{-1}(p)$ , the two-sided level- $\alpha$  test rejecting when the absolute standardized statistic exceeds  $z_{1-\alpha/2}$  has limiting power*

$$1 - \Phi\left(z_{1-\alpha/2} - \frac{\kappa}{\sigma}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{\kappa}{\sigma}\right).$$

*Thus fixed regular directions have the pointwise  $n^{-1/2}$  boundary even though the uniform tie-inclusive boundary in lem:tie-subexperiment-minimax-boundary is slower.*

*Proof.* The target calculation in lem:tie-subexperiment-minimax-boundary gives

$$C_{P_{u_n}}(0, 0) = (b-a)|u_n| = \frac{|\kappa|}{\sqrt{n}}.$$

Let  $G_1 = 1\{S = 1, A = 0\}$  and  $G_0 = 1\{S = 0, A = 0\}$ . Since both cells have probability  $1/4$ ,

$$E_{P_{u_n}} D_i = 4(b-a) \left\{ \frac{1}{4} u_n - \frac{1}{4} \cdot 0 \right\} = \frac{\kappa}{\sqrt{n}}.$$

Because  $Z_i \in \{0, 1\}$ ,  $(Z_i - 1/2)^2 = 1/4$  identically. The cells are disjoint and have total probability  $1/2$ , so

$$E_{P_{u_n}} D_i^2 = 16(b-a)^2 \cdot \frac{1}{2} \cdot \frac{1}{4} = 2(b-a)^2 = \sigma^2.$$

Thus  $\text{Var}_{P_{u_n}}(D_i) \rightarrow \sigma^2$ , and  $|D_i| \leq 2(b-a)$ . For completeness, write  $X_{ni} = (D_i - ED_i)/(\sigma\sqrt{n})$ . Uniform boundedness gives  $\max_i |X_{ni}| \rightarrow 0$ , while  $\sum_i EX_{ni}^2 \rightarrow 1$ . Taylor expansion  $Ee^{itX_{ni}} = 1 - t^2 E[X_{ni}^2]/2 + o(E[X_{ni}^2])$ , uniformly in  $i$ , and multiplication over the independent observations show that the characteristic function of  $\sum_i X_{ni}$  converges to  $e^{-t^2/2}$ . Therefore

$$\frac{\sqrt{n}}{\sigma} \left( \frac{1}{n} \sum_{i=1}^n D_i - \frac{\kappa}{\sqrt{n}} \right) \rightsquigarrow N(0, 1),$$

which is the asserted shifted-normal limit. Taking the probability of the event  $|Z + \kappa/\sigma| > z_{1-\alpha/2}$  for  $Z \sim N(0, 1)$  yields the displayed power formula.  $\square$

**Theorem 1** (thm:tie-local-separation-obstruction). *The frozen assumptions do not imply a uniform root- $n$  local-power conclusion over positive-mass ties. More precisely, they admit the canonical tie subexperiment in lem:tie-subexperiment-minimax-boundary, on which, with  $r_n = n^{-2\beta/(4\beta+d)}$ , every sequence of tests  $\varphi_n$  satisfying*

$$\sup_{P: C_P(0,0)=0} E_P[\varphi_n] \leq \alpha + o(1)$$

*has worst-case power at most  $\alpha + \eta + o(1)$  against some alternatives with  $C_P(0,0) \asymp r_n$ , for every  $\eta > 0$ . There are also alternatives with  $C_P(0,0) \asymp n^{-1/2}$  for which the power is at most  $\alpha + o(1)$ . The rate  $r_n$  is sharp in that subexperiment: the explicit quadratic projection test in lem:tie-subexperiment-minimax-boundary is consistent whenever  $C_P(0,0)/r_n \rightarrow \infty$ . By contrast, on the fixed constant-shift paths of lem:regular-constant-path-power, root- $n$  separation is regular and the two-sided local power is*

$$1 - \Phi\left(z_{1-\alpha/2} - \frac{\kappa}{\sqrt{2}(b-a)}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{\kappa}{\sqrt{2}(b-a)}\right).$$

*Consequently, no uniformly valid test, including an inversion of the frozen numerical directional-delta procedure if it has the claimed level, can have a nontrivial worst-case power curve indexed only by  $\sqrt{n}C_P(\theta)$  over the tie-inclusive primitive class. The frozen conditions therefore answer the local-power part negatively: the regular pointwise boundary is  $n^{-1/2}$ , whereas the uniform positive-mass-tie boundary can be  $n^{-2\beta/(4\beta+d)}$ . This theorem neither proves nor refutes the coverage claim for  $\hat{K}_{n,1-\alpha}$ ; it proves that coverage and level, if established, cannot be supplemented by the requested uniform root- $n$  local-power assertion without restricting the geometry of the alternatives.*

*Proof.* The specialization constructed in lem:tie-subexperiment-minimax-boundary is contained in  $\bar{\mathcal{P}}$ , and  $P_0$  satisfies  $C_{P_0}(0,0) = 0$ . Hence any sequence of tests obeying

$$\sup_{P: C_P(0,0)=0} E_P[\varphi_n] \leq \alpha + o(1)$$

also obeys  $E_{P_0}[\varphi_n] \leq \alpha + o(1)$ . Applying the lower-bound part of lem:tie-subexperiment-minimax-boundary, for every  $\eta > 0$  there is an alternative  $P_{u_n}$  in that subexperiment such that

$$C_{P_{u_n}}(0,0) \asymp c_\eta n^{-2\beta/(4\beta+d)}, \quad E_{P_{u_n}}[\varphi_n] \leq \alpha + \eta + o(1).$$

The same lemma's second mixture choice produces alternatives satisfying  $C_{P_{u_n}}(0,0) \asymp n^{-1/2}$  and  $E_{P_{u_n}}[\varphi_n] \leq \alpha + o(1)$ . Thus a positive-mass tie admits root- $n$  alternatives asymptotically indistinguishable from the null in worst case. Its matching upper-bound calculation shows, within this canonical subexperiment, that the explicit quadratic projection test is consistent whenever

$$\frac{C_{P_u}(0,0)}{n^{-2\beta/(4\beta+d)}} \rightarrow \infty.$$

This establishes the asserted sharp boundary only for the stated subexperiment.

For the contrasting constant directions, `lem:regular-constant-path-power` gives  $C_{P_{un}}(0,0) = |\kappa|/\sqrt{n}$  and a standardized limiting distribution  $N(\kappa/\sigma, 1)$ , where  $\sigma = \sqrt{2(b-a)}$ . Substituting this value of  $\sigma$  into that lemma's two-sided Gaussian rejection probability gives exactly

$$1 - \Phi\left(z_{1-\alpha/2} - \frac{\kappa}{\sqrt{2}(b-a)}\right) + \Phi\left(-z_{1-\alpha/2} - \frac{\kappa}{\sqrt{2}(b-a)}\right).$$

Hence root- $n$  separation is regular along these fixed one-dimensional directions even though it fails uniformly over localized positive-mass-tie directions.

If a uniformly valid test had a nontrivial worst-case power curve determined only by  $\sqrt{n}C_P(\theta)$ , the root- $n$  alternatives from the first paragraph would force that curve to exceed its null level at a fixed positive local separation. Their power is instead at most  $\alpha + o(1)$ , which is a contradiction. This argument applies to any inversion of the numerical directional-delta procedure only conditionally on that inversion having the claimed uniform level. Nothing in either supporting lemma establishes multiplier critical-value validity, confidence-region coverage, or uniform level for that procedure. The coverage and level questions therefore remain undecided, exactly as stated.  $\square$

**Theorem 2** (`thm:exact-compatibility-characterization`). *Suppose `ass:uniform-study-positivity` and `ass:uniform-treatment-positivity` hold. Then, for every  $\theta = (\rho, \gamma) \in [0, 1]^2$ ,*

$$\mathcal{M}(P, \rho, \gamma) \neq \emptyset \iff C_P(\theta) = 0.$$

*Consequently, the exact observed-law compatibility region is the computable zero set*

$$\mathcal{K}(P) = \{\theta \in [0, 1]^2 : \mathcal{M}(P, \rho, \gamma) \neq \emptyset\}.$$

*The equivalence allows arbitrary positive- $P_X$ -mass ties among the clipped lower and upper faces.*

*Proof.* Fix  $\theta = (\rho, \gamma)$ . We prove necessity and sufficiency separately.

*Necessity.* Let  $Q \in \mathcal{M}(P, \rho, \gamma)$ . By the pushforward restriction, on an observational arm with positive conditional probability,

$$E_Q\{Y(t) \mid X, S = 0, A = t\} = E_P\{Y \mid X, S = 0, A = t\} = \mu_{0t}(X).$$

By definition,

$$E_Q\{Y(t) \mid X, S = 0, A = 1 - t\} = c_t(X).$$

Conditioning first on  $A$  therefore gives

$$q_{t0}(X) = e_t(X, 0)\mu_{0t}(X) + e_{1-t}(X, 0)c_t(X) \quad P_{X|S=0}\text{-almost surely.}$$

The study-positivity inequality  $1 - \pi(X) \geq \varepsilon$  makes  $P_{X|S=0}$  and  $P_X$  equivalent. Hence the last display holds  $P_X$ -almost surely. This argument also shows explicitly why division by observational-study mass is required.

The relative-NUC restriction and the nonnegativity of the treatment probabilities imply

$$\begin{aligned} e_t(X, 0)\mu_{0t}(X) + e_{1-t}(X, 0)C_t^-(X; \rho) &\leq q_{t0}(X) \\ &\leq e_t(X, 0)\mu_{0t}(X) + e_{1-t}(X, 0)C_t^+(X; \rho). \end{aligned}$$

By the definition of the exact endpoints, this is

$$q_{t0}(X) \in [W_t^-(X; \rho), W_t^+(X; \rho)] \quad P_X\text{-almost surely.}$$

Study exchangeability also gives

$$q_{t0}(X) \in [V_t^-(X; \gamma), V_t^+(X; \gamma)] \quad P_X\text{-almost surely.}$$

Thus the two closed intervals intersect almost surely, so

$$\max\{V_t^-(X; \gamma), W_t^-(X; \rho)\} \leq \min\{V_t^+(X; \gamma), W_t^+(X; \rho)\}$$

for each  $t$ . Every positive-part violation is zero almost surely, and therefore  $C_P(\theta) = 0$ .

*Sufficiency.* Suppose  $C_P(\theta) = 0$ , and put

$$\ell_t(x) = \max\{V_t^-(x; \gamma), W_t^-(x; \rho)\}, \quad u_t(x) = \min\{V_t^+(x; \gamma), W_t^+(x; \rho)\}.$$

For  $D_t(X) = (\ell_t(X) - u_t(X))_+$ , the definition of  $C_P$  yields  $E_{P_X} D_t = 0$  for every  $t$ , because all of these expectations are nonnegative and their finite maximum is zero. Hence  $D_t = 0$  almost surely. On the resulting common full-measure set, define

$$q_t^*(x) = \frac{\ell_t(x) + u_t(x)}{2}.$$

Then

$$q_t^*(x) \in [V_t^-(x; \gamma), V_t^+(x; \gamma)] \cap [W_t^-(x; \rho), W_t^+(x; \rho)].$$

Since  $e_{1-t}(x, 0) \geq \varepsilon$ , set

$$c_t^*(x) = \frac{q_t^*(x) - e_t(x, 0)\mu_{0t}(x)}{e_{1-t}(x, 0)}.$$

Membership of  $q_t^*$  in the  $W_t$ -interval and division by the positive coefficient  $e_{1-t}(x, 0)$  give

$$C_t^-(x; \rho) \leq c_t^*(x) \leq C_t^+(x; \rho).$$

In particular,  $c_t^*(x) \in [a, b]$ . All displayed selectors are measurable because they are obtained from measurable nuisance versions by finitely many maxima, minima, sums, and division by a function bounded away from zero. Modify them on the exceptional  $P_X$ -null set, if necessary, to take values in  $[a, b]$ .

It remains to construct one common latent law rather than merely compatible conditional means. By Lemma `lem:standard-borel-conditional-kernel`, choose measurable kernels  $\kappa_{st}(x, dy)$  that are versions of the conditional law of  $Y$  given  $X = x, S = s, A = t$ . Positivity ensures

$$\int y \kappa_{st}(x, dy) = \mu_{st}(x) \quad P_X\text{-almost surely.}$$

Write  $p_1(x) = \pi(x)$  and  $p_0(x) = 1 - \pi(x)$ . The observed law then has the disintegration

$$P(dx, \{s\}, \{k\}, dy) = P_X(dx) p_s(x) e_k(x, s) \kappa_{sk}(x, dy).$$

For  $m \in [a, b]$ , the endpoint kernel satisfies

$$\int y \nu_m(dy) = a \frac{b-m}{b-a} + b \frac{m-a}{b-a} = m.$$

Moreover,  $m \mapsto \nu_m(B)$  is Borel measurable for every Borel  $B \subseteq [a, b]$ , so it is a probability kernel. Define conditional laws of the potential-outcome pair by

$$\Lambda_{1k}(x, dy_0, dy_1) = \kappa_{10}(x, dy_0) \kappa_{11}(x, dy_1), \quad k \in \{0, 1\},$$

$$\Lambda_{00}(x, dy_0, dy_1) = \kappa_{00}(x, dy_0) \nu_{c_1^*(x)}(dy_1),$$

$$\Lambda_{01}(x, dy_0, dy_1) = \nu_{c_0^*(x)}(dy_0) \kappa_{01}(x, dy_1).$$

These are measurable probability kernels. Now define  $Q^*$  on  $(X, S, A, Y(0), Y(1))$  by

$$Q^*(dx, \{s\}, \{k\}, dy_0, dy_1) = P_X(dx) p_s(x) e_k(x, s) \Lambda_{sk}(x, dy_0, dy_1).$$

For every  $s, k$ , the  $k$ -th marginal of  $\Lambda_{sk}$  is  $\kappa_{sk}$ . Consequently, the law of  $(X, S, A, Y(A))$  under  $Q^*$  is exactly  $P$ , proving the pushforward restriction. In study one,  $\Lambda_{1k} = \kappa_{10} \otimes \kappa_{11}$  does not depend on  $k$ ; hence

$$(Y(0), Y(1)) \perp_{Q^*} A \mid (X, S = 1).$$

In study zero, treatment positivity and the mean identity for  $\nu_m$  give

$$E_{Q^*}\{Y(t) \mid X, S = 0, A = 1 - t\} = c_t^*(X).$$

Thus relative NUC holds. A direct conditioning calculation gives

$$\begin{aligned} E_{Q^*}\{Y(t) \mid X, S = 0\} &= e_t(X, 0)\mu_{0t}(X) + e_{1-t}(X, 0)c_t^*(X) \\ &= q_t^*(X), \end{aligned}$$

and membership of  $q_t^*$  in the  $V_t$ -interval proves study exchangeability. Hence  $Q^* \in \mathcal{M}(P, \rho, \gamma)$ , establishing sufficiency.

The two directions prove the equivalence. The displayed equality for  $\mathcal{K}(P)$  follows from its definition as the zero set of  $C_P$ . At no step was a unique active face required, so endpoint ties, lower-maximum ties, upper-minimum ties, and outer-arm ties are all included.  $\square$

*Justification.* The theorem upgrades the nonnegative violation functional from a one-sided incompatibility certificate to an exact, computable characterization by constructing a common latent extension whenever every clipped interval intersects. *Gap.* Necessity alone would leave open whether pointwise interval intersection can be realized jointly by measurable potential-outcome kernels. *Consumer.* It identifies the population compatibility frontier as the zero set  $\mathcal{K}(P) = \{\theta : C_P(\theta) = 0\}$ , including positive-mass ties.

**Lemma 5** (lem:scalar-face-tangent-nonfactorization). *In the canonical positive-mass-tie subexperiment of lem:tie-subexperiment-minimax-boundary, let  $P_{\tau,r}$  agree with  $P_0$  except that  $\mu_{10,\tau}(x) = m_0 + \tau r(x)$ , where  $r$  is bounded and belongs to the admitted  $C^\beta$ -ball. For all sufficiently small  $|\tau|$ ,*

$$C_{P_{\tau,r}}(0, 0) = |\tau| \int_{[0,1]^d} |r(x)| dx.$$

*Nevertheless, the two directions  $r_0(x) = 0$  and  $r_1(x) = x_1 - 1/2$  have the same zero derivative at  $\tau = 0$  for every integrated exact-face coordinate  $E[V_t^-]$ ,  $E[V_t^+]$ ,  $E[W_t^-]$ , and  $E[W_t^+]$ , whereas*

$$\left. \frac{d}{d\tau} C_{P_{\tau,r_0}}(0, 0) \right|_{\tau=0+} = 0, \quad \left. \frac{d}{d\tau} C_{P_{\tau,r_1}}(0, 0) \right|_{\tau=0+} = \frac{1}{4}.$$

*Hence the directional derivative of the compatibility functional at this positive-mass tie does not factor through the vector of integrated face derivatives. In particular, the scalar face-score multiplier vector displayed in def:inference-handle does not by itself specify the selector-complete, covariate-indexed tangent needed to differentiate the pointwise max-min and positive-part operations before integration.*

*Proof.* Let  $P_0$  be the null law in the canonical subexperiment and fix  $\theta_0 = (0, 0)$ . The baseline mean  $m_0 = (a + b)/2$  is in the interior of the outcome support. Both  $r_0$  and  $r_1$  are bounded smooth functions, with  $|r_1| \leq 1/2$ . Therefore, for all sufficiently small  $|\tau|$ , the perturbed mean  $m_0 + \tau r_j(x)$  remains in  $[a, b]$ , the corresponding endpoint outcome kernel is valid, and the perturbed regression remains in the fixed Hölder submodel admitted by lem:tie-subexperiment-minimax-boundary. Thus  $P_{\tau,r_j} \in \mathcal{P}$ .

Write  $u_{\tau,r}(x) = \tau r(x)/(b - a)$ . This is exactly the parametrization of the trial cell  $S = 1, A = 0$  used in lem:tie-subexperiment-minimax-boundary, since its conditional outcome mean is

$$m_0 + (b - a)u_{\tau,r}(x) = m_0 + \tau r(x).$$

The target identity proved in that lemma therefore gives

$$C_{P_{\tau,r}}(0, 0) = (b - a)\|u_{\tau,r}\|_1 = |\tau| \int_{[0,1]^d} |r(x)| dx.$$

Taking the right derivative at zero yields

$$\dot{C}_0(r) := \left. \frac{d}{d\tau} C_{P_{\tau,r}}(0, 0) \right|_{\tau=0+} = \int_{[0,1]^d} |r(x)| dx.$$

Consequently  $\dot{C}_0(r_0) = 0$ . Since  $X$  is uniform on  $[0, 1]^d$ , Fubini's identity gives

$$\dot{C}_0(r_1) = \int_0^1 |z - 1/2| dz = 2 \int_0^{1/2} (1/2 - z) dz = \frac{1}{4}.$$

It remains to compare the information retained by the integrated face coordinates. At  $\rho = \gamma = 0$ , clipping is inactive in this interior submodel, and def:exact-endpoints gives

$$V_t^- = V_t^+ = \mu_{1t}, \quad W_t^- = W_t^+ = \mu_{0t}.$$

Along  $P_{\tau,r}$ , only  $\mu_{10}$  changes. Hence the only possibly nonzero integrated face derivatives are those of  $E[V_0^-]$  and  $E[V_0^+]$ , and both equal

$$\left. \frac{d}{d\tau} E_{P_X}[m_0 + \tau r(X)] \right|_{\tau=0} = E_{P_X}[r(X)].$$

For  $r_0$  this expectation is zero. For  $r_1(x) = x_1 - 1/2$ , it is also zero because  $X_1$  is uniform on  $[0, 1]$ . Every other integrated face derivative is identically zero along both paths. Thus the two paths have the same integrated face-derivative vector but distinct compatibility derivatives. No map whose argument is only that vector can represent  $\dot{C}_0$ .

Finally, def:inference-handle defines each  $\psi_h(O; \theta)$  as a scalar score and  $\mathbb{G}_n^\xi = n^{-1/2} \sum_i \xi_i \widehat{\psi}_i$  as the resulting scalar vector over faces and sensitivity indices. The preceding non-factorization shows algebraically that this vector alone omits the covariate-indexed direction on which the pointwise positive part depends. Retaining all scalar face ties does not restore the lost function  $x \mapsto r(x)$ . This proves the lemma.  $\square$

## 7 Interpretation

For each treatment arm,  $C_P(\theta)$  averages the nonnegative amount by which the larger support-clipped lower face exceeds the smaller support-clipped upper face and then takes the maximum across arms. Thus  $C_P(\theta) = 0$  is equivalent to almost-sure conditional interval intersection, not merely nonnegative average signed overlap. The measurable-selector construction turns those intersections into one common latent law, so  $\mathcal{K}(P)$  is an exact compatibility region rather than an outer bound. In the cancellation witness, the stratum weights  $4/5$  and  $1/5$  cancel the signed arm-zero overlaps but leave  $C_{P^{\text{can}}}(1, 1) = 9/10$ . The inference conclusion has two scales: fixed smooth directions are regular at  $n^{-1/2}$ , while uniformly varying positive-mass-tie directions can require  $n^{-2\beta/(4\beta+d)}$ . This proved separation result does not certify the proposed bootstrap’s coverage or level.

## Technical internal limitation (diagnostic only)

The exact compatibility characterization assumes two studies, binary treatment, a known strictly positive bounded outcome support, and uniform study and treatment positivity. Its sensitivity endpoints are specifically the support-clipped Lanners-compatible linear  $(1 \pm \rho)$  and  $(1 \pm \gamma)$  forms. The minimax rate is sharp for the displayed canonical Hölder tie subexperiment; it yields an obstruction to a primitive-class-wide root- $n$  power curve, not a complete minimax theory at every point of the two-parameter frontier. A proof of uniform tie-inclusive coverage and uniform level for the frozen numerical directional-delta construction is still absent.

## 8 Honest scope

The proved results are the observational mixture identity, positive-violation infeasibility, exact common-latent compatibility characterization, the explicit two-stratum cancellation calculation with weights  $4/5$  and  $1/5$  and  $C_{P^{\text{can}}}(1, 1) = 9/10$ , the sharp tie-subexperiment separation rate  $n^{-2\beta/(4\beta+d)}$ , and the regular constant-path Gaussian power formula. The paper does not claim that the proposed cross-fitted numerical directional-delta confidence region has uniform coverage or that its inverted tests have uniform level; those questions remain open. It makes no general optimality claim outside the stated tie subexperiment and no claim beyond the frozen binary-treatment, two-study sensitivity model.

## References

- [1] R. Yu, B. Karmakar, J. Vandelee, and E. B. Schwarz. “Using a Two-Parameter Sensitivity Analysis Framework to Efficiently Combine Randomized and Non-randomized Studies.” arXiv:2412.03731, 2024.
- [2] Q. Lanners, C. Rudin, A. Volfovsky, and H. Parikh. “Data Fusion for Partial Identification of Causal Effects.” *Advances in Neural Information Processing Systems*, 38:150638–150702, 2025. arXiv:2505.24296.

- [3] H. Parikh, M. Morucci, V. Orlandi, S. Roy, C. Rudin, and A. Volfovsky. “A Double Machine Learning Approach for Combining Experimental and Observational Studies.” *arXiv:2307.01449*, 2023.
- [4] V. Chernozhukov, S. Lee, and A. M. Rosen. “Intersection Bounds: Estimation and Inference.” *Econometrica*, 81(2):667–737, 2013.
- [5] M. A. Masten and A. Poirier. “Inference on Breakdown Frontiers.” *Quantitative Economics*, 11(1):41–111, 2020.
- [6] Z. Fang and A. Santos. “Inference on Directionally Differentiable Functions.” *Review of Economic Studies*, 86(1):377–412, 2019.
- [7] H. Hong and J. Li. “The Numerical Delta Method.” *Journal of Econometrics*, 206(2):379–394, 2018.
- [8] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins. “Double/Debiased Machine Learning for Treatment and Causal Parameters.” *Econometrics Journal*, 21(1):C1–C68, 2018.
- [9] A. R. Luedtke and M. J. van der Laan. “Statistical Inference for the Mean Outcome under a Possibly Non-Unique Optimal Treatment Strategy.” *Annals of Statistics*, 44(2):713–742, 2016.
- [10] O. Kallenberg. *Foundations of Modern Probability*. Springer, 2002.
- [11] W. K. Newey. “Convergence Rates and Asymptotic Normality for Series Estimators.” *Journal of Econometrics*, 79(1):147–168, 1997.